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ADAPTIVE FOCUS FOR ADVANCED DIRECTED ENERGY DEPOSITION

Abstract

An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of an additively-manufactured component being manufactured to form a melt pool in the build surface of the component. The system further includes a powder delivery device, a melt pool monitor configured to observe the melt pool, and a computing device. The computing device is configured to receive, from the melt pool monitor, data indicative of one or more parameters of the melt pool and determine, based on the received data, a current position of the melt pool. The computing device is configured to determine a desired size of the melt pool based on the current position of the melt pool and control, based on the desired size of the melt pool, the energy delivery device to form the melt pool of the desired size in the build surface of the component.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to additive manufacturing techniques.

BACKGROUND

[0002] Additive manufacturing generates three-dimensional structures through addition of material layer-by-layer or volume-by-volume to form the structure, rather than removing material from an existing component to generate the three-dimensional structure. Additive manufacturing may be advantageous in many situations, such as rapid prototyping, forming components with complex three-dimensional structures, or the like. In some examples, additive manufacturing may utilize powdered materials and may melt or sinter the powdered material together in predetermined shapes to form the three-dimensional structures.

SUMMARY

[0003] Additive manufacturing systems and techniques, such as directed energy deposition (DED) processes, operate by depositing material layer-by-layer to form an additively-manufactured component (hereinafter, “component”). In such processes, material is deposited by a powder delivery device directing a powder stream toward a melt pool formed on a build surface of the component by energy from an energy delivery device. The melt pool and added powder then solidify as an added layer. The added layer may become the new build surface for subsequent layers. In some examples, the powder delivery device and the energy delivery device may be parts of a common deposition head. The deposition head may travel along a toolpath to deposit a layer. In some examples, the toolpath may track back and forth across the build surface to deposit the layer. It may be beneficial to the performance and life of the component to uniformly apply the deposited layer. In some examples, for instance based on the geometry of the part, it may be desirable to vary the size of the melt pool to deposit a track of consistent thickness, microstructure, porosity, or the like.

[0004] Additive manufacturing systems and techniques according to the present disclosure may allow the size of the melt pool may be varied along a toolpath to deposit a track of material that varies in width. The size of the melt pool may be modified by changing the focus or working distance of the energy delivery device that forms the melt pool. The track of varying width may be uniformly and consistently applied as a layer by automatic modification of the power of the energy delivery device as the melt pool size changes, e.g., to maintain an energy density of the energy delivery device such that the melt pool, which has changed in size, may be kept at a substantially constant depth, temperature, or the like. In some examples, the powder delivery device may be modified such that powder from the powder stream may be added appropriately to a melt pool of changing size. In this way, a substantially uniform layer may be deposited on a complex geometry of the component in a single track, improving uniformity in composition and thickness of the resulting additively-manufactured component.

[0005] An example additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of an additively-manufactured component being manufactured to form a melt pool in the build surface of the component and a powder delivery device configured to direct a powder stream toward the melt pool. The additive manufacturing system also includes a melt pool monitor configured to observe the melt pool and a computing device. The computing

device is configured to receive, from the melt pool monitor, data indicative of one or more parameters of the melt pool. The computing device is configured to determine, based on the received data, a current position of the melt pool. Based on the current position of the melt pool, the computing device may determine a desired size of the melt pool and control, based on the desired size of the melt pool, the energy delivery device to form the melt pool of the desired size in the build surface of the component.

[0006] An example additive manufacturing system includes delivering, via an energy delivery device, energy to a build surface of an additively-manufactured component being manufactured to form a melt pool in the build surface of the component. The technique includes delivering, via a powder delivery device, a powder stream to the melt pool to add material to the component. The technique also includes receiving, from a melt pool monitor, data indicative of one or more parameters of the melt pool. The technique includes determining, based on the received data, a current position of the melt pool. The technique includes determining, via the computing device, based on the position of the melt pool, a desired size of the melt pool. The technique includes controlling, via the computing device, based on the desired size of the melt pool, the energy delivery device to form the melt pool of the desired size in the build surface of the component.

[0007] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 is a conceptual block diagram illustrating aspects of an example additive manufacturing system that includes a powder delivery device configured to direct a powder stream toward a melt pool formed by an energy delivery device.

[0009] FIG. 2 is a conceptual and schematic diagram illustrating further aspects of the additive manufacturing system of FIG. 1.

[0010] FIG. 3 is a conceptual block diagram illustrating portions of the example additive manufacturing system of FIG. 1.

[0011] FIG. 4 is a conceptual block diagram illustrating further aspects of the example additive manufacturing system of FIG. 1.

[0012] FIG. 5 is a process flow diagram illustrating a mass flux monitoring, heat flux monitoring, and melt pool monitoring and control technique.

[0013] FIG. 6 is a conceptual and schematic diagram illustrating operation of a deposition head during an additive manufacturing process.

[0014] FIG. 7 is a conceptual and schematic diagram illustrating operation of the deposition head of FIG. 6 at another point in time during the additive manufacturing process.

[0015] FIGS. 8A-8D are conceptual and schematic diagrams illustrating a deposition head depositing a track of varying width on a component from an isometric view.

[0016] FIGS. 9A and 9B are conceptual views of an example additively-manufactured component from a side and top view, respectively.

[0017] FIGS. 10A-10C are conceptual and schematic diagrams illustrating a deposition head depositing a layer of material on a build surface that includes a deviation.

[0018] FIG. 11 is a flow diagram illustrating an example technique according to the present disclosure.

DETAILED DESCRIPTION

[0019] The disclosure generally describes techniques and systems for depositing a track of varying width during an additive manufacturing technique, such as a directed energy deposition (DED)

technique. During DED additive manufacturing, a component is built up by adding material to the component in sequential layers. The final component is composed of a plurality of layers of material. In some blown powder additive manufacturing techniques for forming a component or a coating on a component from metals, ceramics, alloys, or combinations thereof, an energy source may direct energy at a substrate to form a melt pool. A powder delivery device may deliver a powder to the melt pool, where at least some of the powder at least partially melts and is joined to the melt pool and, thus, the substrate. In some examples, the powder delivery device and the energy delivery device may each be parts of a common deposition head configured to deliver both powder and energy to the build surface. In such cases, the deposition head may travel along a toolpath in a pattern to deposit a layer, for example, by passing back and forth across the build surface, with each pass laying down a track of deposited material.

[0020] The properties of the final component, including the presence or absence of material defects (e.g., pores or void spaces, thickness of each of the plurality of layers, material not joined to adjacent material, and the like), and the efficiency with which material is added to the component, may be a function of the interaction between the powder stream, the melt pool, and the component. As such, measurement, modeling, control, and validation of the position of the powder stream from the powder delivery device, the melt pool formed by the energy delivery device, and the component may enable characterization or prediction of final component properties, control of the additive manufacturing technique during the process, quality assurance for the final component, development of new additive manufacturing techniques, and the like.

[0021] Challenges may arise while performing additive manufacturing techniques with additive manufacturing systems. For example, the component may define a complex geometry. For example, when forming a gas turbine engine component such as an airfoil blade by additive manufacturing, the build surface may vary in width, and at certain points the build surface may be narrow relative to other portions of the build surface. Deposition of a track of material on such a narrow portion of the build surface may require a melt pool that is small in size (e.g., diameter), such that the melt pool does not cause melting over the edge of the build surface. Conversely, however, deposition of a narrow track of material on areas of the build surface which are less narrow may lead to an inefficient additive manufacturing process, since additional passes of the deposition head over the build surface, with each pass laying down a narrow track, may be required to deposit the entire layer. Furthermore, adjacent tracks of material may differ slightly between each other in, for example, in or more of a thickness, microstructure, porosity, or the like. Where possible, it may be desirable to apply a layer on the build surface as a single track to avoid such differences. Accordingly, in addition to gains in efficiency, deposition of a layer in a single pass of a deposition head may improve uniformity of the deposited layer.

[0022] In accordance with one or more examples of the current disclosure, additive manufacturing systems and techniques may address these and other challenges. Additive manufacturing systems according to the present disclosure may include an energy delivery device configured to deliver energy to a build surface of an additively-manufactured component to form a melt pool in the build surface of the component and a powder delivery device configured to direct a powder stream toward the melt pool. The system may include a powder flow monitoring system (PFMS) configured to observe the powder stream and a melt pool monitor configured to observe the melt pool. The system may also include a computing device. The computing device may receive, from melt pool monitor, data indicative of one or more parameters of the melt pool. The computing device may determine, based on the received data, a position and/or a current size of the melt pool. The computing device may determine, based on the position melt pool, a desired size of the melt pool. The computing device may control, based on the determined desired size of the melt pool, the energy delivery device to form the melt pool of the desired size in the build surface of the component. In some examples, controlling the energy delivery device to form the melt pool of the desired size in the build surface may include modifying the energy delivery device to adjust the

size of the melt pool.

[0023] Adjusting the size of the melt pool formed in the build surface may result in deposition of a track or tracks of varying width on the component. Such variation may advantageously allow for formation of narrow geometric features on the component because addition of material to a relatively small melt pool on the component may deposit a narrow track of material, and thus may build a correspondingly narrow feature as the added layer. Additionally, or alternatively, varying the size of the melt pool may increase the efficiency of manufacture by allowing for a reduced number of tracks in relatively wide areas of the build surface through formation of a relatively large melt pool on the build surface. Still further, variation in the size of the melt pool may improve uniformity of the resulting component because variation in the size of the melt pool may allow for a single pass of the deposition head to deposit an entire layer, even when the build surface varies in width, which may improve conformity of the thickness, microstructure, porosity, composition, or the like of the deposited layer.

[0024] The computing device may automatically vary the melt pool size in-situ during the additive manufacturing process, which may improve the efficiency and quality of the process relative to additive manufacturing systems which must be stopped and/or manually adjusted to vary the melt pool size. For example, the energy delivery device may be a laser which delivers an energy beam which impinges on the build surface to form the melt pool. The size of the melt pool may be defined by the greatest dimension of the melt pool in a plane parallel to the build surface. Typically, the energy beam and the melt pool have a substantially elliptical cross-sectional area in a plane parallel to the build surface such that the greatest dimension of the melt pool is the melt pool diameter, but other shapes are also considered. For example, one or more shields or other occlusion devices may be employed to block some of the energy beam to change the shape of the melt pool, and the size of the melt pool may be defined by the greatest dimension of the resulting melt pool. The melt pool may be varied in size by modification of the focus of the laser (e.g., by adjustment of a lens and/or mirror system) such that the spot size of the laser on the build surface may be made larger or smaller. The melt pool may form in the build surface at the laser spot. Additionally, or alternatively, modification of the working distance, which is the distance between the energy delivery device and the build surface, may be employed to vary the melt pool size, in addition to or alternatively to modification of the focus.

[0025] To vary the melt pool size, the computing system may control the energy delivery device to form the melt pool of the desired size in the build surface. The desired melt pool size may be determined in any suitable way. For example, to determine the desired size of the melt pool, the computing device may store a model of a final fully-formed version of the additively-manufactured component (e.g., finished component). The model of the fully-formed additively-manufactured component may be divided into a matrix with a plurality of cells. For example, the matrix may include a plurality of rows, with each row corresponding to a layer of the plurality of layers of the component and plurality of columns, with each column corresponding to a portion of a plurality of portions of the component. The model may store desired melt pool sizes at each cell of the matrix, along with other aspects of the build strategy (e.g., system settings such as power supplied to the energy delivery device, mass flux, powder density, carrier gas flow rate, etc.). The computing system may determine the current position of the melt pool, determined based on received data from a melt pool monitor, and may compare the current position of the melt pool to the model of the fully formed additively-manufactured component to determine the location of the melt pool within the model, that is, which cell in the matrix contains the desired melt pool size corresponding to the particular position. The computing device may thus determine the desired size of the melt pool based at partially on the model of the fully formed version of the additively-manufactured component.

[0026] Other ways to determine the desired size of the melt pool are considered. For example, in some examples, a track may be positioned at an edge of the component, and the additive

manufacturing system may deposit a track that aligns with the edge, even when the complex geometry of the component requires that the track vary in width. In some examples, the computing device may receive, from the melt pool monitor, data indicative of the geometry of the build surface (e.g., edges of the build surface) in addition to one or more parameters of the melt pool. The computing device may determine at least one boundary of the build surface based on the received data, and determine the desired size of the melt pool based at least partially on the determined boundary or boundaries. In this way, by sensing an edge or edges of the component, the additive manufacturing system may deposit a layer with the same or similar dimensions as the underlying layer without causing melting over the edge of the layer by modifying the size of the melt pool, and thus the resulting track of deposited material, to conform to the edge of the component.

[0027] Still other ways to determine the desired size of the melt pool are considered. For example, a topology sensor, which in some cases may be a laser profilometer, may scan the build surface and output data indicative of the topology of the build surface in the area of the melt pool. The computing device may receive the topological data from the topology sensor and identify, based on the received data, a deviation in the build surface. For example, the received topological data may be compared to specification data representative of a set of tolerances of the build surface to identify the deviation. The deviation may be a depression or crack in the build surface, or may be a relatively high area such as a ridge or protrusion. The deviation may result from melt pool splatter, unmelted powder from the powder stream adhering to the build surface, or the like. The computing device may control the energy delivery device to modify the size of the melt pool in the vicinity of the build surface where the deviation is recognized, based on an algorithm, look-up-table, machine learning model, or the like. The algorithm, look-up-table, or machine learning model may store or output a desired melt pool size for the identified deviation.

[0028] In some examples, machine learning may be used to identify the correct or target melt pool size at various positions within the volume of the build for given materials, component geometry and set of deposition parameters. By producing a known build of good quality, the melt pool shape presented to the sensing device and then collecting analyzing the images, a point-to-point and/or moment-to-moment target melt pool size can be derived from the data.

[0029] In cases where a machine learning model is used, the machine learning model may be trained on data from previously identified deviations and/or corrective actions. For example, a protrusion cause by unmelted powder at the build surface may be identified and at least partially repaired by increasing the size of the melt pool at or near the protrusion, which may melt the unmelted powder and/or flatten the build surface in the area of the protrusion. In this way, aspects of the current disclosure may allow for in-situ monitoring and corrective action of the additive manufacturing process where the melt pool may be varied in size to improve the quality of the resulting component.

[0030] To control the energy delivery device to form the melt pool of the desired size, the system may include logic for deciding whether an adjustment to the melt pool size is necessary. For example, the computing device may determine whether the current size of the melt pool meets a threshold for matching the desired size of the melt pool. Responsive to determining that the current size of the melt pool does not meet the threshold for matching the desired size of the melt pool, the computing device may adjust the size of the melt pool by modifying the energy delivery device (e.g., adjust the focus or working distance of the energy delivery device). In some examples, the melt pool may be substantially elliptical or circular, and may vary in diameter from about 0.1 millimeters (mm) to about 5 mm, such as from about 0.4 mm to about 3 mm. In some examples, the threshold for matching the desired size of the melt pool may be about plus or minus (+/-) 5 percent (%), or about 10%, or about 20%.

[0031] Varying the melt pool in size during the deposition may allow for varying the width of the deposited track, but if other settings of the system are not modified (e.g., proportionally modified),

the track may vary in porosity, composition, microstructure, or the like as the track varies in width. It may be desirable, for increased quality of the final component, to deposit a track that is uniform in parameters other than the varying width. In conventional systems, many parameters are not manipulable during operation of the additive manufacturing system, or may be tied to other settings such that changing one setting forces other settings to change. For example, some additive manufacturing systems require that a convergence point of the powder stream change if the melt pool size changes, which may not be desirable in all circumstances. Aspects of the current disclosure relate to additive manufacturing systems and techniques that allow for independent manipulation of settings under control of the computing device. Decoupling, for example, the powder delivery device and the energy delivery device such that each device is independently controllable may allow for selective tailoring of process settings under control of the computing device. In this way, the computing device may control the settings and operation of the energy delivery device and the powder delivery device to respond appropriately to a melt pool that varies in size.

[0032] For example, if the powder stream is directed in the same volume and direction at a melt pool that varies between relatively small and relatively large sizes, powder from the powder stream may be added to a different point in the melt pool at a different volume or may miss the melt pool altogether, reducing the capture efficiency of the melt pool. It may be desirable to add powder from the powder stream to the melt pool in a way that varies according to the changing size of the melt pool. For example, the powder delivery device may include one or more delivery nozzles from which the powder stream is directed, and the computing device may control a position of the one or more delivery nozzles to change the way powder is added to the melt pool as the melt pool varies in size. In some examples, the powder delivery device includes a plurality of delivery nozzles where individual portions of the powder stream are directed from each delivery nozzle of the plurality of delivery nozzles. The individual portions of the powder stream may be configured to converge at a convergence point. The computing device may, responsive to a change in the size of the melt pool, adjust the powder delivery device (e.g., an angle of the delivery nozzles) to modify the convergence point. In this way, the powder stream may be controlled as the melt pool size varies to maintain capture efficiency, component properties, or the like.

[0033] Varying the melt pool size without varying other settings of the energy delivery device may be problematic. For example, if the melt pool is made larger or smaller without varying the power supplied to the energy delivery device, the energy density of the energy beam forming the melt pool will increase or decrease. The energy density may be considered the units of energy impinging upon the build surface divided by the area of the melt pool. Changing the energy density of the energy beam supplied to the melt pool may deleteriously impact the temperature of the melt pool and ultimately the way in which material is added to the melt pool and cooled to form a new layer.

[0034] According to aspects of the current disclosure, the computing device may maintain a power density of the energy delivery device by adjusting the power supplied to the melt pool proportionally to the adjustment of the size of the melt pool. For example, to maintain the energy density of the melt pool, the computing device may increase the power supplied to the energy delivery device as the melt pool grows in size and reduce the power supplied to the melt pool as the melt pool shrinks in size. In this way, material may be added and joined to melt pools of relatively smaller and relatively larger size in a uniform manner, and narrow geometric features of the deposited layer and larger areas of the layer may share final material properties. Although primarily discussed herein as maintaining the energy density, in some examples the computing device may increase or decrease the energy density of the energy beam responsive to a change in the size of the melt pool.

[0035] One or more aspects of the present disclosure may be used to account for the way an edge or a topological deviation cause the melt pool to appear to the melt pool monitor. As deposition moves over the build surface, the melt pool size and geometry will change depending on the

location of the melt pool to the underlying shape. At edges or corners, as the melt pool rolls over the edge, the melt pool may appear smaller to the camera than in the center section of the build. The melt pool will also appear differently to the camera as it travels up or down an incline or over a wavy surface. For example, the size and/or shape of the melt pool may distort when the melt pool is positioned on or near a topological deviation in the build surface. According to aspects of the present disclosure, the desired size of the appearance of the melt pool to the melt pool monitor may be determined, based on the underlying geometry of the component. In some examples, the computing device may store a model of the actual size of the melt pool based on the predicted distortion or disappearance of the melt pool by the melt pool monitor. In some examples, data captured by the melt pool monitor during deposition on one layer in real time and then feeding back that information into the deposition of the subsequent layer will allow for a more accurate assessment and choosing of the correct target melt pool size.

[0036] FIG. 1 is a conceptual block diagram illustrating aspects of an example additive manufacturing system 10. Additive manufacturing system 10 includes several components configured to monitor mass flow, and several components configured to monitor energy (e.g., heat flux) within system 10. System 10 includes a powder source mass sensor 44, a powder flow monitoring system (PFMS) 18, and a topology sensor 48. These components are configured to monitor mass flow of powder within additive manufacturing system 10 during an additive manufacturing technique. System 10 also includes melt pool monitor 15, and includes other components which measure and monitor heat flux within system 10, which are further described with respect to FIG. 3. As such, to simplify illustration of FIG. 1 and improve clarity of the figure, further aspects of additive manufacturing system 10 are shown in FIG. 3 and described below with reference to FIG. 3, which is more directed toward heat flow aspects of system 10. In the example illustrated in FIG. 1, additive manufacturing system 10 further includes a computing device 12, a powder delivery device 14, an energy delivery device 16, a stage 20, a powder source 42, powder source mass sensor 44, and topology sensor 48. Computing device 12 is operably connected to powder delivery device 14, energy delivery device 16, PFMS 18, stage 20, powder source 42, powder source mass sensor 44, and topology sensor 48. FIG. 1 thus illustrates mass flow monitoring and other aspects of example additive manufacturing system 10.

[0037] Stage 20 is configured to mechanically support a component 22 during an additive manufacturing technique. Component 22 may be considered in-situ when mechanically supported by stage 20. In some examples, stage 20 is movable relative to energy delivery device 16 and/or energy delivery device 16 is movable relative to stage 20. Similarly, stage 20 may be movable relative to powder delivery device 14 and/or powder delivery device 14 may be movable relative to stage 20. For example, stage 20 may be translatable and/or rotatable along at least one axis to position component 22 relative to energy delivery device 16 and/or powder delivery device 14. Similarly, energy delivery device 16 and/or powder delivery device 14 may be translatable and/or rotatable along at least one axis to position energy delivery device 16 and/or powder delivery device 14, respectively, relative to component 22. Stage 20 may be configured to selectively position and restrain component 22 in place relative to stage 20 during manufacturing of component 22.

[0038] Powder source 42 is the source of powder for powder stream 30. Powder source 42 may include any suitable container or enclosure, such as a hopper, configured to hold powder. Powder source 42 also may include mechanism for entraining the powder in a gas flow. For instance, powder source 42 may be coupled to a gas source, which provides a gas flowing through powder source 42 and entraining powder within the gas flow. Additionally, or alternatively, powder source 42 may include an agitator configured to agitate the powder and increase entrainment of the powder in the gas stream.

[0039] System 10 may include a powder source mass sensor 44 associated with powder source 42. Powder source mass sensor 44 may be configured to quantify loss of mass in the powder source 42

or, alternatively, a mass flow out of powder source **42**.

[0040] Powder source **42** is fluidically coupled to powder delivery device **14** via a flow path **46**. Flow path **46** may include any suitable structure(s) defining an enclosed flow between powder source **42** and powder delivery device, including conduit, pipe, tubes, or the like. Although not shown in FIG. **1**, for at least part of flow path **46** between powder source **42** and nozzles of powder delivery device **14**, flow path **46** may split into multiple, parallel sections, e.g., one for each nozzle. Further, although not shown in FIG. **1**, in some examples, flow path **46** may include one or more nozzles for controlling flow through flow path **46** as a whole or portions of flow path **46** (e.g., a section associated with a particular nozzle of powder delivery device **14**).

[0041] Powder delivery device **14** may be configured to deliver powder to selected locations of component **22** being formed via a powder stream **30**. Powder delivery device **14** may include one or more delivery nozzles that each output a portion powder as powder streams **30A** and **30B**. Portions **30A** and **30B** may converge at a convergence point (e.g., at build surface **28**) to define powder stream **30**. In some examples, where the convergence point is below build surface **28**, portions **30A** and **30B** of powder stream **30** may not converge, and rather enter melt pool **32** at different points before having a chance to converge. In some examples, powder delivery device **14** includes a single delivery nozzle, which may be point nozzle, or a single delivery nozzle that is an annular channel. In other examples, powder delivery device **14** includes a plurality of delivery nozzles (e.g., three nozzles or four nozzles). Regardless of the number of delivery nozzles, powder delivery device **14** may output a powder stream that is focused at a focus plane, which may also be called a convergence point. The convergence point may be the point at which powder streams **30A** and **30B** meet to form powder stream **30**. As powder delivery device **14** is movable in the z-axis shown in FIG. **1** relative to component **22**, the convergence point of powder stream **30** also may be movable in the z-axis relative to component **22**, such that the focus plane may be controlled to be substantially coincident with build surface **28**. In some examples, the convergence point may be below build surface **28**, such that powder streams **30A** and **30B** are added to melt pool **32** as separate powder streams. In some examples, the convergence point of powder stream **30** may be manipulated even without moving powder delivery device **14** in the z-direction, such as by adjusting the angle of the delivery nozzles of powder delivery device **14**.

[0042] At least some of the powder in powder stream **30** may impact a melt pool **32** in component **22**. At least some of the powder that impacts melt pool **32** may be joined to component **22**. In some examples, powder delivery device **14** may be mechanically coupled or attached to energy delivery device **16** to facilitate delivery of powder stream **30** and energy beam **34** for forming melt pool **32** to substantially the same location adjacent to component **22**. As will be further described below, the size of melt pool **32**, which may be defined as the greatest dimension in a plane parallel to build surface **28** at the center of melt pool **32**, may be adjustable by control of energy delivery device **16**. For example, increasing the focus of energy beam **34** such that it defines a smaller cross-sectional area may cause melt pool **32** to shrink in size, while reducing the focus of energy beam **34** may cause melt pool **32** to enlarge in size. In some examples, reducing the working distance, or distance from energy delivery device **16** may cause melt pool **32** to shrink in size, and increasing the working distance may cause melt pool **32** to enlarge in size.

[0043] Energy delivery device **16** may include an energy source, such as a laser source, an electron beam source, plasma source, or another source of energy that may be absorbed by component **22** to form a melt pool **32** and/or be absorbed by powder in powder stream **30** to be added to component **22**. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, the energy source may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique.

[0044] In some examples, energy delivery device **16** also includes an energy delivery head, which is operatively connected to the energy source. The energy delivery head may aim, focus, or direct

energy beam **34** toward predetermined positions at or adjacent to build surface **28** of component **22** during the additive manufacturing technique. As described above, in some examples, the energy delivery head may be movable in at least one dimension (e.g., translatable and/or rotatable) under control of computing device **12** to direct energy **34** toward a selected location at or adjacent to build surface **28** of component **22**. Energy delivery device **16** may be configured to focus energy **34** from the energy source on a local spot on build surface **28** to generate melt pool **32**. As will be further described below, energy delivery device **16** may include one or more mechanisms to adjust the size of energy beam **34** and thus melt pool **32**. For example, energy delivery device **16** may direct energy **34** through an optical system that includes one or more mirrors and/or one or more lenses, which may be manipulated to manipulate the focus of energy beam **34**. Additionally, or alternatively, energy delivery device **16** may be mounted on one or more adjustable support members, which may be adjusted by computing device **12**. The galvanometers, lenses, and/or adjustable support members may be adjusted by computing device **12** to modify and selectively tailor the size of energy beam **34**, and, accordingly, the position of melt pool **32** on build surface **28**.

[0045] In some examples, at least a portion of energy delivery device **16** and powder delivery device **14** may be combined or attached to each other. For example, a deposition head (e.g., deposition head **54** of FIG. 2) may include part of powder delivery device **14** (e.g., internal channels and powder delivery nozzle(s) **56** for forming powder streams **30A** and **30B**, which converge as powder stream **30**) and part of energy delivery device **16** (e.g., the energy delivery head). As shown in FIG. 1, in some examples, primary energy delivery device **16** may be arranged of configured such that energy **34** and powder stream **30** both exit from a common deposition head (**54**, FIG. 2) and are directed toward build surface **28**. For instance, energy **34** may pass through a central channel (e.g., formed along central longitudinal axis L, FIG. 2) within the deposition head and exit a central aperture in the deposition head, while fluidized powder may flow through internal channels and powder nozzle(s) **56** for forming powder stream **30** and directing powder stream **30** toward build surface **28**. Such an arrangement between primary energy source **16** and powder delivery device **14** may be called an “on-axis” arrangement of energy source **16**, because both energy and powder may be delivered coaxially with a central longitudinal (Z-direction) axis of the deposition head.

[0046] System **10** also includes powder flow monitoring system (PFMS) **18**. PFMS **18** is configured to image at least a portion of powder stream **30** to detect powder flowing between powder delivery device **14** and build surface **28**. For example, PFMS **18** may include an illumination device and an imaging device. In some examples, the illumination device may include one or more light sources. For instance, the illumination device may include one or more structured light devices, such as one or more lasers. The illumination device is configured to illuminate a plane of powder stream **30** at image plane **38**, e.g., a plane substantially perpendicular to an axis extending between powder delivery device **14** and build surface **28** (e.g., central longitudinal axis L).

[0047] The imaging device of PFMS **18** is configured to image at least some of the illuminated powder. The imaging device may have a relatively high data acquisition speed (e.g., frame rate), such greater than 1000 Hz. Because of the velocity of the powder in powder stream **30**, even such a frame rate may image only a fraction of the powder flowing between powder delivery device **14** and build surface **28**. In some examples, the imaging device may be a camera.

[0048] In some examples, PFMS **18** also includes a housing configured to enclose the illumination device and the imaging device. The housing may be configured to protect the illumination device and the imaging device from damage due to the harsh conditions to which PFMS **18** may be exposed during use. For example, the housing may protect the illumination device and the imaging device from powder deflections from powder stream **30** off build surface **28**, may cool the illumination device and the imaging device to remove heat incident on PFMS **18** from melt pool **32**

and energy delivery device **16**, or the like.

[0049] PFMS **18** may be positionally fixed relative to powder delivery device **14** and/or energy delivery device **16**, e.g., in the x-y plane shown in FIG. **1**. This may help maintain a relative x-y position of PFMS **18** and the image plane of the imaging device relative to powder stream **30**. This may facilitate analysis of image data captured by the imaging device.

[0050] PFMS **18** may be movable in the z-axis direction of FIG. **1** (e.g., parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28**). This may enable movement of image plane **38** along the z-axis of FIG. **1** (e.g., parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28**). This may allow PFMS **18** to image powder stream **30** at different positions between powder delivery device **14** and build surface **28**. In this way, PFMS **18** may analyze powder stream **30** along powder stream **30** to help determine parameters of powder stream **30** along its length.

[0051] In some example, PFMS **18** may be positionally fixed relative to powder delivery device **14** and/or energy delivery device **16** and movable parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28** by an adjustable z-stage **40**. Adjustable z-stage **40** may be attached to energy delivery device **16**, powder delivery device **14**, or a portion of system **10** that moves energy delivery device **16** and/or powder delivery device **14**, such that PFMS **18** moves in the x-y axis in registration with energy delivery device **16** and/or powder delivery device **14**.

[0052] Adjustable z-stage **40** may be controlled by computing device **12** to position PFMS **18** and image plane **38** relative to powder stream **30**. Further, computing device **12** may control adjustable z-stage **40** to move PFMS **18** vertically and out of the way to allow powder delivery device **16** and energy delivery device **16** access to physically constrained areas, e.g., between vanes of a doublet or triplet of a nozzle guide vane for a gas turbine engine.

[0053] System **10** further includes a topology sensor **48**. Topology sensor **48** is configured to monitor an amount of powder captured by melt pool **32** by imaging melt pool **32** and the added material, allowing the mass to be quantified (e.g., by computing device **12**) using the dimensions of the added material and density of the material (powder). In some examples, topology sensor **48** includes a laser and a sensor (e.g., an imaging device, such as a camera), which senses laser light reflected by the structure being imaged (e.g., melt pool **32** and the added material). Such a device may be called a laser profilometer. The laser may have a defined wavelength, which may affect the resolution of the topology sensor **48**. In some examples, the wavelength and sensor may be selected such that the resolution of topology sensor **48** is as great as about 10 microns (e.g., about 6 microns).

[0054] In some examples, topology sensor **48** may be positioned substantially directly above component **22** (e.g., along a central axis of powder delivery device **14**, energy delivery device **16**, or a common deposition head that includes both powder delivery device **14** and energy delivery device **16**) and may include an interferometer, which provides depth information based on the time from outputting a laser pulse to the sensing of the reflected light. In other examples, topology sensor **48** may be positioned at an offset with respect to component **22** such that the sensor senses depth information without using an interferometer.

[0055] In some examples, topology sensor **48** may be integral with system **10**, e.g., disposed within the enclosure or working area of system **10**. In other examples, topology sensor **48** may be an add-on component to system **10**. For example, the enclosure in which the additive manufacturing technique is performed may include a transparent window, and topology sensor **48** may be positioned outside of the enclosure and may image component **22** through the transparent window.

[0056] Although a topology sensor **48** is described in the examples of this disclosure, in other examples, another metrology device may be utilized to determine the amount of powder captured by melt pool **32**. For example, another type of light source may be used. In some examples, if another type of light source is used, component **22** or stage **20** may include one or more features that serve as indicators of scale. Furthermore, although described as a single topology sensor **48**,

more than one sensor may be used, and may employ more than one of the technologies described above.

[0057] Computing device **12** may control components of system **10** and may include, for example, a desktop computer, a laptop computer, a workstation, a server, a mainframe, a cloud computing system, or the like. Computing device **12** may control operation of system **10**, including, for example, powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, topology sensor **48**, and/or further elements of system **10**, such as melt pool (MP) monitor **15**. Computing device **12** may be communicatively coupled to powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, MP monitor **15**, and/or topology sensor **48** using respective communication connections. In some examples, the communication connections may include network links, such as Ethernet, ATM, or other network connections. Such connections may be wireless and/or wired connections. In other examples, the communication connections may include other types of device connections, such as USB, IEEE 1394, or the like.

[0058] Computing device **12** may include one or more processors. Example of processors include, but are not limited to, one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. Computing device **12** may include a memory, and the memory may store instructions for various operations of system **10**.

[0059] Although FIG. **1** illustrates a single computing device **12** and attributes all control and processing functions to that single computing device **12**, in other examples, system **10** may include multiple computing devices **12**, e.g., a plurality of computing devices **12**. In general, control and processing functions described herein may be divided among one or more computing devices. For instance, system **10** may include a controller for energy delivery device **16**, powder delivery device **14**, and stage **20**, a separate controller for PFMS **18**, and a separate computing device for analyzing data obtained by PFMS **18**, mass sensor **44**, MP monitor **15**, and topology sensor **48**. As another example, system may include a dedicated controller for each of energy delivery device **16**, powder delivery device **14**, stage **20**, PFMS **18**, MP monitor **15**, and topology sensor **48**, and a separate computing device for coordinating control of powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48** and analyzing data obtained by PFMS **18**, MP monitor **15**, powder source mass sensor **44**, and/or topology sensor **48**. Other examples of computing system architectures for controlling system **10** and analyzing data obtained from system **10** will be apparent and are within the scope of this disclosure.

[0060] Computing device **12** may control operation of powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40**, stage **20**, MP monitor **15**, and/or topology sensor **48** to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MP monitor **15**, and topology sensor **48**. For example, as described above, computing device **12** may control stage **20** and powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40**, and/or topology sensor **48**, to translate and/or rotate along at least one axis to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MP monitor **15**, and topology sensor **48**. Positioning component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MP monitor **15**, and topology sensor **48** may include positioning a predetermined surface (e.g., build surface **28**, a surface to which material is to be added) of component **22** in a predetermined orientation relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology sensor **48**.

[0061] Computing device **12** may control system **10** to deposit layers **24** and **26** to form component body **25** and eventually finished component **22**. As shown in FIG. **1**, component **22** may include a first layer **24** and a second layer **26**, although many components may be formed of additional

layers, such as tens of layers, hundreds of layers, thousands of layers, or the like. Component **22** in FIG. **1** is simplified in geometry and the number of layers compared to many components formed using additive manufacturing techniques, as described and illustrated with respect to FIGS. **8A-8D** below. Although techniques are described herein with respect to component **22** including first layer **24** and second layer **26**, the technique may be extended to components **22** with more complex geometry and any number of layers. Furthermore, although component **22** is illustrated as being uniform, in some examples component **22** may be functionally-graded and include at least two different portions having different selectively tailored properties.

[0062] To form component **22**, computing device **12** may control powder delivery device **14** and energy delivery device **16** to form, on build surface **28** of first layer **24** of material, a second layer **26** of material using an additive manufacturing technique. Computing device **12** may control energy delivery device **16** to deliver energy beam **34** to a volume at or near surface **28** to form melt pool **32**. For example, computing device **12** may control the relative position of energy delivery device **16** and stage **20** to direct energy to the volume. Computing device **12** also may control powder delivery device **14** to deliver powder stream **30** to melt pool **32**. For example, computing device **12** may control the relative position of powder delivery device **14** and stage **20** to direct powder stream **30** at or on to melt pool **32**. The position of powder stream **30** and melt pool **32** may be independently controlled by computing device **12**, such that the relative position of powder stream **30** to melt pool **32** may be selectively controlled as part of an additive manufacturing technique, in accordance with examples of this disclosure. The relative position of powder stream **30** to melt pool **32** may even be independently controlled when powder delivery device **14** and energy delivery device **16** are parts of a common deposition head (**54**, FIG. **2**). Such independent control of powder delivery device **14** and energy delivery device **16** during an additive manufacturing process may permit adjustment of the relative position of powder stream **30** to melt pool **32** to account for settings and parameters of system **10** during operation. Computing device **12** may control energy delivery device **16** to cause a change in the size of melt pool **32**, such as by a modification of the focus or z-axis position of energy delivery device **16**.

[0063] Computing device **12** may then control a z-axis position of stage **20** and/or powder delivery device **14** and energy delivery device **16** such that melt pool **32** will be formed on surface **36** of second layer **26**, and may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. Each individual pass in the pattern may be called a track, such that layer **26** is made up of a plurality of adjacent tracks. Computing device **12** may control powder delivery device **14** and energy delivery device **16** similarly until all layers are formed to define a completed component **22**.

[0064] Computing device **12** may control powder delivery device **14** and primary energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. As mentioned above, powder delivery device **14** and energy delivery device **16** may be parts of a common deposition head, and computing device **12** may control the deposition head (**54**, FIG. **2**) to move along a toolpath to deposit layer **26**. Computing device **12** may receive, from PFMS **18**, data indicative of the position of powder stream **30**. Computing device **12** may also receive, from MP monitor **15**, which is configured to observe the melt pool, data indicative of a current position and size of melt pool **32** on build surface **28**. The current position of melt pool **32** may be defined relative to common deposition head (**54**, FIG. **2**), which may be defined within system **10**. Additionally, or alternatively, the current position of melt pool **32** be defined with respect to a fully formed version e.g., a point cloud model of dimensions, of final component **22**. In other words, computing device **12** may store a model of the desired finished component **22**. The model may be a matrix with a plurality of rows, each row representing a layer **26**, and a plurality of columns, each column representing a portion of component **22**. Computing device **12** may store, at each location in the matrix, a build strategy for component **22**. The build strategy may include various operational settings of system **10**. The build strategy may include a desired size of melt

pool 32, and may include settings of energy delivery device 12 (e.g., focus settings, power supplied, etc.) to achieve the desired size of melt pool 32. In this way, the desired size of melt pool 32 may be determined based on the location of melt pool 32 within the model of the desired final component 22.

[0065] In some examples, computing device 12 may store/execute one or more machine learning models which may be used to adapt control of powder delivery device 14, primary energy delivery device 16, or another component. For example, data captured by MP monitor 15, energy delivery device 16, PFMS 18, topology sensor 48, mass source 44, or other components of system 10 may be used to train the machine learning device. Furthermore, additional parameters of system 10 may be inputs to a machine learning model, such as operational settings of system 10.

[0066] In some examples, the machine learning model stored and executed by computing device 12 machine learning may be used to identify the correct or target melt pool size at various positions within the volume of the build. For example, the machine learning model may include inputs that include the given materials, component 22 geometry and deposition parameters of system 10. The size of melt pool 32 may also be an input to the machine learning model. By producing a known build of good quality, the melt pool shape presented to the sensing device and then collecting analyzing the images, a point-to-point and/or moment-to-moment target melt pool size can be derived from the data.

[0067] The machine learning model may also be used to account for the way an edge or a topological deviation in component 22 may cause melt pool 32 to appear in the data captured by melt pool monitor 15. For example, as deposition moves over build surface 28, the melt pool size and geometry of melt pool 32 may change depending on the location of melt pool 32 and the relationship of melt pool 32 to the underlying shape of component 22. At edges or corners, or over topological defects (e.g., protrusions or depressions on build surface 28, and or over concave or convex portions of build surface 28, melt pool 15 may capture data that indicates melt pool 32 as changing in size or shape.

[0068] For example, as melt pool 32 rolls over an edge, melt pool 32 may appear smaller to melt pool monitor 15 than the same melt pool would appear if positioned in a center section of build surface 28. Similarly, melt pool 32 may also appear differently to melt pool monitor 32 as it travels up or down an incline or over a wavy portion of build surface 28. In some examples, data captured by melt pool monitor 15 during deposition may be input into the machine learning model. The machine learning model may compare, from the same or previous additive manufacturing processes, and using data input that includes the geometry of component 22 and settings of powder delivery device 14 and energy delivery device 16 the appearance of the melt pool and output a target size of melt pool 32. In this way, by capturing and analyzing data collected during deposition of one layer in real time and then feeding back that information into the deposition of the subsequent layer, system 10 may allow for a more accurate assessment and choosing of the correct target size of melt pool 32.

[0069] FIG. 2 is a conceptual and schematic diagram illustrating example additive manufacturing system 100. Additive manufacturing system 100 of FIG. 2 may be an example of additive manufacturing system 10 of FIG. 1. System 100 includes an example powder flow monitoring system 50 configured to monitor powder flow between a powder delivery device 52 and a build surface (not shown in FIG. 2) during an additive manufacturing technique. Powder delivery device 52 may be an example of powder delivery device 14 of FIG. 1, and PFMS 50 may be an example of PFMS 18 of FIG. 1.

[0070] Powder delivery device 52 includes a deposition head 54 that carries a plurality of delivery nozzles 56. Plurality of delivery nozzles 56 output a powder stream 58 toward the build surface. As shown in FIG. 2, the powder stream 58 may be focused at convergence point CP, such that powder stream 58 is converging toward the focal plane and diverging away from the focal plane. As discussed above, deposition head 54 may further include energy delivery device 16, which is not

illustrated in FIG. 2 for improved clarity. Powder delivery device 52 may define powder stream central axis L. In examples where powder delivery device 52 includes a single delivery nozzle 56, central axis L may pass through the single delivery nozzle. In some examples, as illustrated, powder delivery device 52 may include plurality of delivery nozzles 56, and powder stream central axis L may be defined as passing through a point equidistant from each of the plurality of delivery nozzle EP. Point EP may be the point equidistant from plurality of delivery nozzles 56 on the downstream end, relative to powder flowing through powder delivery device 52, of powder delivery device 52. Point CP may be the point at which powder from the plurality of delivery nozzles 56 converges, which may be at the build surface (28, FIG. 1) of a component. In some examples, the energy delivery device (not illustrated in FIG. 2 for clarity), may be configured to deliver energy to the build surface parallel to (e.g., coincident with) powder stream central axis L. [0071] PFMS 18 includes a housing 60 (also referred to as an enclosure), which encloses an imaging device 62 and an illumination device 64. In some examples, imaging device 62 may be a high-speed camera and illumination device 64 may be laser illuminator. Housing 60 is attached to an adjustable z-stage 66 by a bracket 68.

[0072] Housing 60 may enclose imaging device 62 and illumination device 64 and help protect imaging device 62 and illumination device 64 from a surrounding environment. For instance, housing 60 may surround imaging device 62 and illumination device 64 and prevent any powder that reflects from the build surface toward PFMS 18 from impacting imaging device 62 or illumination device 64.

[0073] Further, housing 60 may be configured to cool imaging device 62 and illumination device 64. Imaging device 62 and illumination device 64 may be exposed to heat from the melt pool at the build surface and energy from the energy delivery device. Imaging device 62 and illumination device 64 may be relatively sensitive to heat and have improved operational lifetime if maintained and operated below a certain temperature. PFMS 50 may include a cooling system 70 that removes heat from within housing 60 to cooling imaging device 62 and illumination device 64. For instance, cooling system 70 may include cooling fluid circuit through which a cooling fluid flows, and housing 60 may include part of the cooling circuit. In some examples, housing 60 may be formed from a material having relatively high thermal conductivity, such as aluminum, to help transfer heat from within housing 60 to cooling system 70 (e.g., a cooling fluid flowing through cooling system 70).

[0074] As described above, PFMS 50 may be configured to measure powder flow of powder stream 58 at one or more axial (or longitudinal) locations of powder stream 58 and determine one or more parameters associated with the powder flow. For instance, PFMS 50 may be determine a position of powder stream, either absolutely with respect to system 100 or with respect to deposition head 54, the position of which may be known relative to system 100. For instance, illumination device 64 may illuminate powder of powder stream 58 in a plane oriented substantially orthogonal to a longitudinal axis that extends from powder delivery device 52 to the build surface. PFMS 50 may be positioned at a selected axial or longitudinal location to image a selected axial or longitudinal position between powder delivery device 52 and the build surface. Imaging device 62 may be a camera configured to image at least some of the illuminated powder. PFMS 50 may include a computing device (e.g., computing device 12 of FIG. 1) configured to analyze images captured by imaging device 62 to identify a number of particle detections in each captured image and, optionally, derive further parameters from the number of particle detections. As such, computing device 12 may receive image data representing an image captured by imaging device 62. The image data may include representations of illuminated powder of powder stream 58, as imaged by imaging device 62 (e.g., as captured in an image frame by imaging device 62). Computing device 12 may generate a representation of powder stream based on the image data and output the representation of the powder stream for display at a display device.

[0075] In some examples, computing device 12 may determine a powder mass flow represented by

the image data. To do so, computing device **12** may identify a number of powder particles within each image frame. In some examples, computing device **12** additionally may identify a size and/or shape of each powder particle within each image frame. Computing device **12** may implement any suitable image analysis technique to identify powder particles, and, optionally, size and/or shape of powder particles. Since the position of imaging device **62**, either absolutely within additive manufacturing system **100** or relatively with respect to associated deposition head **54**, may be known, the position of powder stream **58** may be determined by the position of powder stream **58** within the image frame.

[0076] Once computing device **12** has identified a number of powder particles within an image frame, computing device **12** may determine a mass flow based on the number of powder particles. For example, computing device **12** may determine the mass flow based on a calibration equation or calibration curve. The relationship between particle detections and mass flow may be determined experimentally. For instance, the relationship between particle detections and mass flow may be determined for each powder type (e.g., composition, size distribution, or both), as each powder type may have a different relationship between particle detections and mass flow. The relationship may be determined experimentally by flowing a known mass of powder at a known rate, and imaging the powder. By doing this multiple times at multiple rates, the calibration curve may be generated. The curve, in the form of an equation, a look-up table, or the like, may be stored in computing device **12**, and computing device **12** may use the calibration curve to determine mass flow of a similar type of powder at a different flow rate based on particle detections.

[0077] In some examples, computing device **12** may receive image data representative of a sequence of images of illuminated powder in powder stream **58**. Each image may be associated with a time. As such, computing device **12** may select one or more images of the sequence of images and analyze the one or more images. For each selected image, computing device **12** may identify a number of particle detections and, optionally, determine a mass flow associated with powder stream **58** for each image frame.

[0078] As described above, system **10** may include both mass flow monitoring and heat flow monitoring. FIG. **1** best illustrates the mass flow monitoring aspects of system **10**. FIG. **3** is a conceptual block diagram illustrating further aspects of system **10**, best illustrating the heat flow monitoring aspects of the example system. System **10** includes an optical system **80** for observing thermal emissions around melt pool **32**, MP monitor **15** which includes a thermal camera for monitoring one or more parameters (e.g., a size, temperature, or the like) of melt pool **32**, and an optional off-axis camera **21**. Identical reference numerals in FIGS. **1** and **3** refer to the same parts. Further, those common parts are the same or substantially identical, aside from any differences described herein. As shown in FIG. **3**, energy delivery device **16** includes an optical system **80**.

[0079] During additive manufacturing, component **22** is built up by adding material to component **22** in sequential layers. The final component is composed of a plurality of layers of material. Energy delivery device **16** may direct energy beam **34** at build surface **28** of first layer **24** to form melt pool **32**. Powder delivery device **14** may deliver powder stream **30** to melt pool **32**, where at least some of the powder at least partially melts and is joined to first layer **24**. Melt pool **32** cools as energy beam **34** is no longer delivered to that location of first layer **24** (e.g., due to energy delivery device **16** scanning energy **34** over the surface of first layer **24**). The temperature and cooling rate of melt pool **32** and the surrounding areas of first layer **24** affect the microstructure, porosity, temper, and other properties of component **22** formed using the additive manufacturing technique. Further, these and other properties of component **22** may be impacted by the size of melt pool **32**. Optical system **80**, MP monitor **15**, and/or camera **21** may capture data indicative of parameters of melt pool **32** (e.g., a position such as a position of a center point of melt pool **32** and/or a size of melt pool **32**), and computing device **12** may analyze the captured data and determine the position of melt pool **32**, either absolutely within system **10** or relatively with respect to an associated deposition head (e.g., deposition head **54**, FIG. **2**).

[0080] Optical system **80** may include an imaging device and an associated optical train, which senses emissions at or near component **22** during the additive manufacturing technique. For example, optical system **80** may include a visible light imaging device, an infrared imaging device, or an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range.

[0081] The optical train may include one or more reflective, refractive, diffractive optical components configured to direct light to the imaging device. For example, the optical train may be configured to direct light from near component **22** and/or melt pool **32** to the imaging device. In some examples, at least a portion of the optical train is coaxial with the axis at which energy delivery device **16** outputs energy, and the at least a portion of the optical train may be attached to or otherwise configured to move with the portion of energy delivery device **16** that directs or focuses energy **34** at or near the surface of component **22**. In this way, optical system **80** may move with energy delivery device **16** and track melt pool **32** as melt pool **32** moves across component **22**, without needing to correct for any offsets between energy delivery device **16** and optical system **80** and/or needing to correct for geometry of component **22**. In other examples, the optical train may not be coaxial with the axis at which energy delivery device **16** outputs energy **34**, and computing device **12** may be configured to compensate for the offset and any affects this may have on the imaging, including shadowing, interference, geometry of component **22**, or the like. Off-axis camera **21**, which is a second camera not coaxial with the axis at which energy delivery device **16** outputs energy beam **34**. Off-axis camera **21** may be configured to determine the position of melt pool **32** within system **10** by imaging energy **34** and/or melt pool **32** from a known (e.g., fixed) position within system **10**.

[0082] In some examples, optical system **80** may include an occulting device. The occulting device is configured to reduce or block emissions (e.g., thermal emissions) that originate from the energy output by energy delivery device **16** and/or near a center of melt pool **32**, which otherwise obfuscate emissions from solidifying regions of material at or near the edge of melt pool **32** and outside of melt pool **32**. The occulting device may be a rigid occulting device or a dynamic occulting device. A rigid occulting device reduces or blocks emissions from a fixed region, e.g., from the energy **34** output by energy delivery device **16**. For instance, a rigid occulting device may include a device with fixed dimensions that is opaque to wavelengths of interest. As another example, a rigid occulting device may include an apodizing lens in which a center of the lens is substantially opaque to wavelengths of interest and opacity decreases as a function of radius.

[0083] A dynamic occulting device is configured to be controlled to occult different regions, e.g., different sizes and/or shapes. A dynamic occulting device may include a rigid occulting device that is mounted to a device that can translate the rigid occulting device along and/or perpendicularly to the optical axis. As another example, a dynamic occulting device may include an opaque and viscous liquid, such as mercury, contained between two substrates. The substrates are substantially transparent to the wavelength(s) of interest. One or both of the substrates may be movable relative to the other substrate to control the distance between the substrates. By reducing the distance between the substrates, the size of the occulting region may increase. By increasing the distance between the substrates, the size of the occulting region may decrease. As a third example, a dynamic occulting device may include a digital micromirror device. Computing device **12** may control the micromirrors of the digital micromirror device to direct emissions that originate from energy **34** output by energy delivery device **16** and/or near a center of the melt pool away from the imaging device. A digital micromirror device may enable control of both the size and shape of the region of emissions that are occulted. As will be further discussed below, computing device **12** may a digital micromirror device within energy delivery device **16**, which may be a galvanometer, or another digital micromirror device, to adjust the position of energy **34**, and thus the position of melt pool **32**.

[0084] FIG. **4** is a conceptual block diagram illustrating an example optical system **80** for

observing thermal emissions at and/or around a melt pool **32** and or a focused spot on a build surface **28** where a melt pool is not formed during an additive manufacturing technique. Optical system **80** includes an optical train that includes first imaging optics **92**, occulting device **94**, second imaging optics **96**, and imaging device **98**. Imaging device **98** may be any suitable imaging device, including, for example, a visible light imaging device, an infrared imaging device, an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range, a two color pyrometry imaging device, or the like.

[0085] First and second imaging optics **92** and **96** may each include one or more optical devices used to direct light to imaging device **98**. For example, First and second imaging optics **92** and **96** may each include one or more refractive optical device (e.g., a lens), one or more reflective optical device (e.g., a mirror), one or more diffractive optical devices (e.g., a grating), one or more dichroic optical devices (e.g., a dichroic filter or mirror), or the like. Although two sets of imaging optics **92** and **98** are shown in FIG. **4**, in other examples, system **80** may include a single set of imaging optics or more than two sets of imaging optics.

[0086] Occulting device **94** is positioned within the optical train between first imaging optics **92** and second imaging optics **96**. In other example, occulting device **94** may be positioned between imaging device **98** and imaging optics **96** or after before imaging optics **92**. In some examples, occulting device **94** is positioned as the optical component nearest imaging device. This effectively results in removal of the portion of the image which occulting device **94** blocks. In other examples, occulting device **94** is positioned at another position within the optical train **80** where the image of component **22** resolves. Imaging optics **96** then may be configured to image occulting device **94** onto imaging device **98**.

[0087] As shown in FIG. **4**, in some examples, at least a portion of optical system **80** is coaxial with the axis at which energy delivery device **16** outputs energy **34**. For example, at least a portion of second imaging optics **92** (e.g., the portion at which thermal emissions **104** is incident upon second imaging optics **92**) may be coaxial with the axis at which energy delivery device **16** outputs energy **34**. This may reduce image manipulation that otherwise may be applied to the resulting image to correct for geometry of component **22**, angular offset of optical system **80** relative to energy delivery device **16**, shadowing due to the angular offset, interference, or the like. In other examples, optical system **80** (e.g., the portion at which thermal emissions **104** are incident upon second imaging optics **92**) may not be coaxial with the axis at which energy delivery device **16** outputs energy **34**, and computing device **12** (FIG. **1**) or another computing device may manipulate the resulting image to compensate for geometry of component **22**, angular offset of optical system **80** relative to energy delivery device **16**, shadowing due to the angular offset, interference, or the like.

[0088] FIG. **4** also illustrates energy delivery device **16** outputting energy **34**, which is incident upon component **22** and results in formation of melt pool **32**. Surrounding melt pool is a cooling zone **102**, in which temperature gradients from the temperature of melt pool **32** to ambient temperature are present. As shown in FIG. **4**, melt pool **32** and cooling zone **102** emit thermal emissions **104** (e.g., thermal radiation), which travel through optical system **80** to imaging device **98**, which images the thermal emissions **104**. Occulting device **94** occults (e.g., reduces the intensity of or substantially eliminates) thermal emissions **104** from a selected region, e.g., a region corresponding to energy **34** and at least a portion of melt pool **32**. This may allow imaging device **98** to more effectively image relatively lower intensity thermal emissions from at or near the edge of melt pool **32** and within cooling zone **102**. This may enable more accurate measurement of temperatures within the cooling zone **102**, and heat flow within cooling zone **102**.

[0089] Returning to FIG. **3**, system **10** also includes melt pool monitor (“MP monitor”) **15**. Melt pool monitor **15** may include one or more sensors for monitoring a characteristic of melt pool **32**. The monitored characteristic may be indicative of a temperature of melt pool **32**. For example, the one or more sensors of MP monitor **15** may include an imaging system, such as a visible or thermal

camera, e.g., camera to visible light or infrared (IR) radiation. The visible light camera may capture data representative of the geometry of the melt pool and/or the surrounding build surface **28**, e.g., a width, diameter, shape, or the like.

[0090] Computing device **12** may process the captured image data to determine a size and position of melt pool **32**. In some examples, MP monitor **15** may detect the temperature of melt pool **32** at multiple positions within the melt pool, such as a leading edge, a center, and a trailing edge of the melt pool. In some examples, the imaging system may include a relatively high-speed camera capable of capturing image data at a rate of tens or hundreds of frames per second or more, which may facilitate real-time detection of the characteristic of melt pool **32**. In some examples, MP monitor **15** may capture data at a sequence of particular points in time including a first point in time, a second point in time, etc. In some examples, data from melt pool monitor **15** may be analyzed by computing device **12** to determine a current position of melt pool **32** on build surface **28**. For example, computing device **12** may determine a center point of melt pool **32**. As mentioned above, computing device **12** may determine a size of melt pool **32**. In some examples, the determined size may be based on a greatest dimension of melt pool **32** at build surface **28** (e.g., a distance from the leading edge of liquid material to the trailing edge of liquid material, a diameter of a substantially circular melt pool, a corner to corner distance of a rectangular melt pool, or the like).

[0091] FIG. 5 is a process flow diagram illustrating an additive manufacturing monitor and control technique. The technique of FIG. 5 may be implemented by system **10** of FIGS. 1 and 3 and will be described with concurrent reference to FIGS. 1 and 3. However, it will be appreciated that system **10** may perform other techniques and the technique of FIG. 5 may be performed by other systems. For example, system **100** of FIG. 2 may perform the described technique.

[0092] One or more computing devices **12** may be configured to control a powder feed rate output by powder source **42** (see top left of FIG. 5). For instance, one or more computing devices **12** may be configured to control an agitator of powder source **42**, a gas flow rate of gas flowing through powder source **42**, a position of one or more valves within flow path **46**, or the like to control a powder feed rate output by powder source **42**.

[0093] One or more computing devices **12** may be configured to receive data from one or more mass flow monitoring sensors, including PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. Data received from powder flow mass sensor **44** indicates a mass flow of powder from powder source **42** to powder delivery device. Data from PFMS **18** indicates a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**. Data from topology sensor **48** indicates powder mass captured by melt pool **32** and added to component **22**.

[0094] One or more computing devices **12** may calculate one or more mass flow-related metrics based on the data received from PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. For example, one or more computing devices **12** may determine a capture efficiency by determining a fraction or percentage of powder from powder stream **30** that is captured by melt pool **32** and added to component **22**, e.g., by dividing the powder mass captured by melt pool **32**, as determined based on data from topology sensor, into the mass flow determined based on data received from PFMS **18**.

[0095] Further, one or more computing devices **12** may determine an overall mass flux using the data received from PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. One or more computing devices **12** then may use the overall mass flux as an input to the control algorithm used to control the powder feed rate output by powder source **42** (see top left of FIG. 5). Additionally, one or more computing devices **12** may determine the position of powder stream **30**, either absolutely within system **10** or relatively with respect to a deposition head.

[0096] Similarly, one or more computing devices **12** may be configured to control energy delivery device **16** to deliver energy beam **34** to first layer **24** to establish a given heat input (see bottom left of FIG. 7). For example, one or more computing device **12** may control one or more operating

parameters of energy delivery device **16**, such as by maintaining or adjusting a spot size (e.g., by maintaining or adjusting a focus or working distance), an intensity, a pulse rate, a pulse width, or the like; one or more positional parameters related to energy delivery device **16**, such as dwell time at a location, a movement rate relative to first layer **24**, an overlap between adjacent passes of energy **34** across first layer **24**, a pause time between adjacent passes of energy **34** across first layer **24**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**).

[0097] One or more computing devices **12** may be configured to receive data captured by one or more thermal sensors, such as optical system **80** and/or melt pool monitor **15**. One or more computing devices **12** may determine a cooling rate and associated heat from using data from optical system **80** and may determine a heat input into component **22** using the determined size and/or temperature of melt pool **32** as observed by melt pool monitor **15**. In some examples, one or more computing devices **12** receive data captured by melt pool monitor **15**, and may analyze the captured data to determine a position of melt pool **32** (e.g., a center point of melt pool **32**).

[0098] In some examples, the technique of FIG. **5** includes validation of the dimensions of melt pool **32**. For example, computing device **12** may determine the current size of melt pool **32** as described above. Based on the location of melt pool **32**, computing device **12** may determine a desired melt pool size. Computing device **12** may control, based on the desired size of melt pool **32**, energy delivery device **16** to form melt pool **32** of the desired size in build surface **28** of component **22**. Computing device **12** may repeat the process of comparing the current size of melt pool **32** to the desired size of melt pool **32** to iterate and validate the formation of the melt pool of the desired size in build surface **28**.

[0099] Computing device **12** may determine the desired size of melt pool **32** in one or more ways. For example, to determine the desired size of melt pool **32**, computing device **12** may store a model of a fully formed version of component **22**. The model of the desired final dimensions of fully formed component **22** may be divided into a matrix with a plurality of cells. For example, the matrix may include a plurality of rows, with each row corresponding to layers **24**, **26** of the plurality of layers of component **22** and a plurality of columns, with each column corresponding to a portion of a plurality of portions of component **22**. Each portion may be defined substantially parallel to the build direction of layers **24**, **26**. The model may store desired melt pool sizes at each cell of the matrix, along with other aspects of the build strategy (e.g., settings of system **10** such as power supplied to energy delivery device **16**, mass flux, powder density, carrier gas flow rate, etc.). Computing device **12** may compare the current position of melt pool **32**, as determined, for example, by captured data from melt pool monitor **15**, to the model of the fully formed component **22** to determine the current location of melt pool **32** within the model, that is, which cell in the matrix contains the desired melt pool size corresponding to the particular current position. Computing device **12** may thus determine the desired size of melt pool **32** based at partially on the model of final component **22**.

[0100] Additionally, or alternatively, as will further described below, in some examples computing device **12** may determine a desired size of melt pool **32** based at least partially on a sensed edge of previously deposited layer **24** of component **22**. For example, a track may be positioned at an edge of component **22**, and system **10** may deposit a track that aligns with the edge, even when the complex geometry of component **22** requires that the track vary in width. In some examples, computing device **12** may receive, from melt pool monitor **15** and/or optical system **80**, data indicative of the geometry of build surface **28**. Computing device **12** may determine at least one boundary (e.g., the edge) of build surface **28** based on the received data from melt pool monitor **15** and optical system **80**, and determine the desired size of melt pool **32** based at least partially on the determined boundary or boundaries. In this way, by sensing an edge or edges of component **22**, system **10** may deposit layer **26** with the same or similar dimensions as underlying layer **24**. Advantageously, melting over the edge of layer **24** may be reduced or eliminated by modifying the size of melt pool **32**, and thus the resulting track of deposited material, to conform to the edge of

component **22**.

[0101] Additionally, or alternatively, as will be further described below, computing device **12** may determine a desired size of melt pool **32** based at least partially on the topology of build surface **28**. For example, topology sensor **48**, which in some cases may be a laser profilometer, may scan build surface **28** and output data indicative of the topology of build surface **28** in the area of melt pool **32**. Computing device **12** may receive the topological data from topology sensor **48** and identify, based on the received data, a deviation in build surface **28**. For example, the received topological data from topology sensor **48** may be compared to specification data representative of a set of tolerances of build surface **28** to identify the deviation. The deviation may be a depression or crack in build surface **28**, or may be a relatively high area such as a ridge or protrusion. The deviation may result from melt pool splatter (e.g., material ejected from melt pool **32**), unmelted powder from powder stream **30** adhering to build surface **28**, or the like. Computing device **12** may control energy delivery device **16** to modify the size of melt pool **32** in the area of build surface **28** where the deviation is recognized, based on an algorithm, look-up-table, machine learning model, or the like. The algorithm, look-up-table, or machine learning model may store or output a desired melt pool size for the identified deviation.

[0102] In cases where a machine learning model is used, the machine learning model may be trained on data from previously identified deviations and/or corrective actions. For example, a protrusion cause by unmelted powder at build surface **28** may be identified and at least partially repaired by increasing the size of melt pool **32** at or near the protrusion, which may melt the unmelted powder and flatten the build surface in the area of the protrusion. In this way, system **10** may determine a desired size of melt pool **32** based on the position of melt pool **32** based on a model of component **22**, a sensed boundary of component **22**, and/or a topology of build surface **28**.

[0103] In some examples, the desired size of melt pool **32** may be determined based on a combination of the above-described techniques. For example, the desired size of melt pool **32** may be initially determined by comparison to the model of the fully formed version of component **22**. The output desired size of melt pool **32** may be modified by a determined boundary of component **22** and/or an identified deviation in build surface **22**. In this way, the desired size of melt pool **32** may be determined by computing device **12** based on a combination of techniques.

[0104] To control the energy delivery device to form melt pool **32** at the desired size, computing device **12** may include logic for deciding whether an adjustment to the current size of melt pool **32** is necessary. For example, computing device **12** may determine whether the current size of melt pool **32** meets a threshold for matching the desired size of melt pool **32**. Responsive to determining that the current size of melt pool **32** does not meet the threshold for matching the desired size of melt pool **32**, computing device **12** may adjust the size of melt pool **32** by modifying energy delivery device **16** (e.g., adjust the focus or working distance of energy delivery device **16**). In some examples, melt pool **32** may be substantially circular, and computing device **12** may vary melt pool **32** in diameter from about 0.1 millimeters (mm) to about 5 mm, such as from about 0.4 mm to about 3 mm. In some examples, the threshold for matching the desired size of the melt pool may be about plus or minus (+/-) 5 percent (%), or about 10%, or about 20%.

[0105] Maintaining or adjusting the current size of melt pool **32** to meet the threshold for matching the desired size of melt pool **32** by control of energy delivery device **16** may be accompanied by manipulation of other settings of system **10** by computing device **12**. For example, computing device **12** may manipulate a power density of energy delivery device **12** by modifying a power supplied to energy delivery device **16**. In some examples, computing device **12** may maintain the energy density supplied to melt pool **32** as the size of melt pool **32** varies. Computing delivery device **12** may additionally or alternatively control powder delivery device **14** based at least partially on the size of melt pool **32**. With reference to FIG. 2, computing device **12** may control a position (e.g., a delivery angle) of delivery nozzles **56**.

[0106] FIGS. 6 and 7 are conceptual and schematic diagrams illustrating a portion of additive manufacturing system **200**. System **200** may be an example of system **10** of FIGS. 1 and 3 or system **100** of FIG. 2. System **200** includes deposition head **254**, which includes both powder delivery device **214** and energy delivery device **216**. Several additional components, including one or more computing devices, cameras, and sensors as described above with respect to system **10** and system **100** are not illustrated in FIGS. 6 and 7 for clarity. Components not illustrated in FIGS. 6 and 7 are described with respect to system **10** of FIGS. 1 and 5.

[0107] Deposition head **254** includes energy delivery device **216**, which may generate and/or delivery energy beam **234** to build surface **228** of component **222** to form melt pool **232**.

Deposition head **254** includes powder delivery device **214**, which directs powder stream **230** to the melt pool. Powder in powder stream **230**, at least some of which may be at least partially melted by energy **234** while in flight across working distance WD, is captured by melt pool **232**, and solidifies to form layer **226** on component **222** during the additive manufacturing process. Computing device **12** may cause deposition head **254** to travel along a toolpath, indicated by arrow T in FIGS. 6-7, and the relative position of powder stream **230** to melt pool **232** may be maintained as deposition head **254** travels along toolpath T to deposit layer **226**.

[0108] Deposition head **254** houses energy delivery device **216**, which in the illustrated example is a laser. Energy delivery device **216** may reside in a cavity within deposition head **254**, and may be supported by one or more adjustable support members **202**. In operation, computing device **12** may control adjustable support members **202** in one or more of the X, Y, or Z directions, via, for example, an electrical connection. Computing device **12** may actuate adjustable support members **202** to adjust the position of energy delivery device **216** within deposition head **254**. In this way, computing device **12** may control the position of melt pool **232** on build surface **228**, and thus may control the spot size of energy beam **234**. In some examples, energy delivery device **216** may include optical train **204**, which may include a plurality of mirrors and/or lenses configured to position and shape energy beam **234**. Adjustment of optical train **204** by computing device **12** may allow for selective tailoring of the spot size, position, frequency, polarity, and/or other aspects of energy beam **234** delivered to build surface **228** to form melt pool **232**.

[0109] Powder travels from powder source **42** through powder delivery device **214** via channels formed in the walls of deposition head **254**, and exits delivery nozzles **256A**, **256B** (collectively, “delivery nozzles **256**”) as powder stream **230A**, **230B** (collectively, “powder stream **230**”). Powder stream **230** traverses working distance WD between delivery nozzles **256** and build surface **228**. Delivery nozzles **256** may be controlled independent of energy delivery device **216** by computing device **12** to adjust the position of powder stream **230** relative to melt pool **232**. For example, delivery nozzles **256** may direct powder stream **230** towards melt pool **232** at an angle, and the angle may be adjusted by computing device **12**. Similarly, the orifice size or position of delivery nozzles **256** may be adjusted by computing device **12**, in examples where powder delivery device **214** allows for these parameters to be adjusted in situ.

[0110] FIG. 6 illustrates deposition of layer **226** by deposition head **254** during a first point in time during an additive manufacturing operation of component **222**, and FIG. 7 illustrates deposition head **254** during a second point in time during the operation. At the first point in time of FIG. 6, energy delivery device **216** forms melt pool **232** at first size S1, and at the second point in time of FIG. 7, energy delivery device **216** forms melt pool **232** at second size S2. As illustrated, sizes S1 and S2 are different from each other. In some examples, as illustrated, computing device **12** may manipulate optical train **204** to adjust size of melt pool **232** between sizes S1 and S2.

[0111] As further illustrated by the broken lines of FIG. 7, responsive to varying the size of melt pool **232** from size S1 of FIG. 6 to size S2 of FIG. 7, computing device **12** may manipulate other variables of system **200**. For example, computing device **12** may adjust the angle of delivery nozzles **256** by angle α . Adjusting the angle of delivery nozzles **256** by angle α may manipulate the convergence point of powder stream **230**. In this way, responsive to the changing size of melt pool

232, computing device **12** may manipulate the addition of powder to melt pool **232**, for example to more evenly distribute powder stream **230** throughout a larger melt pool.

[0112] Computing device **12** may manipulate a power supplied to energy delivery device **216** to control an energy density supplied to melt pool **232**. Energy density is represented in FIGS. **6** and **7** by the density of the broken lines indicating energy beam **234**. Computing device **12** may increase the power supplied to energy delivery device **12** from the first point in time of FIG. **6** to the second point in time of FIG. **7**, responsive to the increased size of melt pool **232**. In some examples, computing device **12** may maintain the energy density of energy beam **234** to melt pool **232** as the melt pool increases in size from size **S1** to size **S2**, as illustrated by the substantially similar density of broken lines.

[0113] FIGS. **8A-8D** are conceptual and schematic diagrams illustrating system **300** during an additive manufacturing operation where layer **326** is deposited on component **322**. System **300** may be an example of system **10** of FIGS. **1** and **3** or system **100** of FIG. **2**, where similar reference numerals indicate similar elements. FIGS. **8A-8D** illustrate deposition of a layer **326**, progressing from before the track begins in FIG. **8A** to completion of the track in FIG. **8D**. Component **322** includes build surface **328**, which varies in width from a relatively wide portion **P1** to a relatively narrow portion **P2**.

[0114] FIG. **8B** illustrates deposition of layer **326** by deposition head **354** as deposition head **354** travels along toolpath **T** in relatively wide portion **P1**. To completely cover build surface **328** in wide portion **P1**, melt pool **332** may be relatively large, defining size **D1**. In some examples, size **D1** may be a diameter, and may be about 3 mm. As used herein, the term about encompasses the stated value and those values within 10% of the stated value. To maintain a proper energy density for the relatively large melt pool, the power supplied to the energy delivery device (not illustrated) may be about 2 kilowatts (kW).

[0115] Deposition head **354** may progress further along toolpath **T** to deposit layer **326** on relatively narrow portion **P1**, as illustrated by FIG. **8C**. To deposit layer **326** on the relatively narrow portion **P1**, computing device **12** may manipulate the energy delivery device to change the size of melt pool **332** such that melt pool **332** defines relatively smaller size **D2**. In some examples, size **D2** may be about 0.4 mm. To maintain final material properties of layer **326**, computing device **12** may reduce the power supplied to the energy delivery device to maintain about the same energy density. In some examples, the energy supplied to the energy delivery device in FIG. **8C** may be about 100 W. Deposition head **354** may continue to progress along the toolpath **T** until completing a single pass across build surface **328**, depositing layer **326** in a single track as illustrated in FIG. **8D**. In this way, by varying the size of melt pool **332**, system **300** may deposit layer **326** as a single track, even where component **322** defines a complex geometry including relatively wide portions **P1** and relatively narrow portions **P2**. Relative to a layer deposited in multiple tracks, layer **326** may define more uniform material properties.

[0116] FIGS. **9A** and **9B** are conceptual views of an example additively-manufactured component **422** from a side and top view, respectively. Component **422** may be additively manufactured with systems and techniques according to the present disclosure. Component **422** includes component body **425** and additively-manufactured coating layer **426** disposed on component body **425**. Layer **426** may be deposited (e.g., by system **10** of FIGS. **1** and **3**) by deposition of tracks **470A-470G** (collectively “tracks **470**”). Tracks **470A-470G** may be sequentially laid down by passing a deposition head (**54**, FIG. **2**) back and forth over component body **425**.

[0117] In some examples, as illustrated, component **422** may define a curved edge **472**. Edge **472** may also be called a boundary of component **422**. As such, it may be desirable to deposit track **470A**, which is disposed on edge **472**, with relatively narrow portion **W1** and relatively wide portion **W2**, such that layer **426** conforms to edge **472**. System **10** may deposit track **470A** along edge **472** by varying the size of the melt pool along track **470A**. In this way, both melting over edge **472** and increased overlap with track **470B** may be reduced or avoided, which may increase the

quality of component **422**.

[0118] FIGS. **10A-10C** are conceptual and schematic diagrams illustrating deposition head **554** of system **500** depositing layer **526** to form component **522**. System **500** may be an example of system **10** of FIGS. **1** and **3**, and similar reference numerals may describe similar elements.

[0119] Component **522** includes component body **525** and underlying layer **524** disposed on component body **525**. Deposition head **554** may deposit layer **526** on build surface **528** of underlying layer **524**. Underlying layer **524** includes deviation **580**, which is a topological deviation from the substantially planar build surface **528**.

[0120] Deviation **580** in the illustrated example of FIG. **10A** is a protrusion, but in other examples may be another topological feature such as a depression, crack, ridge or the like. Deviation **580** may be caused by any one or more of a number of events during an additive manufacturing process, such as melt pool splatter, agglomerations of material in powder stream **30**, unmelted powder falling on surface **528**, or the like. In any case, deviation **580** protrudes from the remainder of build surface **528**. Deposition of layer **526** on deviation **580** without addressing or compensating for deviation in layer **526** resulting from deviation **580** may result in final component **522** missing a specification window, reduce proper adherence between layers **524** and **526**, or otherwise cause deleterious effects. As described above, system **500** may include topology sensor **48** and computing device **12**. Topology sensor **48** may capture topological data indicative of deviation **580**, and computing device **12** may identify deviation **580** in the captured topological data. In some examples, computing device **12** may determine that deviation **580** causes layer **524** to fall outside a threshold specification window. Responsive to determining that deviation **580** causes layer **524** to fail to meet a specified tolerance, computing device **12** may determine that the size of the melt pool should be adjusted on or near deviation **580** as a corrective action. For example, computing device **12** may store one or more previous deviations, and may execute an algorithm, a lookup table, and/or a machine learning model trained on successful and/or unsuccessful corrective actions taken on these previous deviations. Based on the identified deviation **580**, computing device **12** may control system **500** to form adjusted melt pool **532B** on surface **528**.

[0121] For example, as illustrated in FIG. **10B**, as deposition head **554** travels along a toolpath, computing device **12** may cause melt pool **332A** to increase in size to define melt pool **332B**. The increased size of melt pool **332B** may at least partially flatten deviation **580**. As a result, layer **526** may be deposited with reduced or eliminated deviations, as illustrated in FIG. **10C**. Accordingly, system **500** may vary the size of the melt pool to improve the topology, and thus the resulting quality, of component **522**.

[0122] FIG. **11** is a flow diagram illustrating an example technique for additive manufacturing a component or coating based on the desired size of the melt pool. The technique of FIG. **11** may be executed by system **10** of FIGS. **1** and **3**, system **100** of FIG. **2**, system **200** of FIGS. **6** and **7**, system **300** of FIGS. **8A-8D**, or system **500** of FIG. **10B**, and will be described as such, although other systems may be used to perform the described technique and other techniques may be performed using the described systems. The technique of FIG. **11** may be performed to additively-manufacture aerospace parts or coatings such as turbine blades or vanes and coatings disposed thereon.

[0123] Energy delivery device **16** may deliver energy beam **34** to build surface **28** of component **22** to form melt pool **32** in build surface **28** (**602**). Component **22** may, in some examples, be a gas turbine engine component. In some examples, component **22** may define a complex geometry not defined by rectilinear polygons, for example relatively narrow features that may, in some cases, be smaller than can be manufactured by traditional additive manufacturing systems. In some cases, energy delivery device may be a laser. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, energy delivery device **16** may both generate and deliver energy **34**. In some examples, energy **34** may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique.

[0124] Powder delivery device **14** may deliver powder stream **30** to melt pool **32** to add material to component **22** as layer **26** (**604**). In some examples, powder stream **30** may be powder entrained in a carrier gas or gases. The powder may include a metal or alloy, such that additively-manufactured component **22** includes a metal or an alloy. In some examples, powder delivery device **14** and energy delivery device **16** may be parts of a common deposition head **54**. Deposition head **54** may travel along toolpath **T** to deposit layer **26** of material on a component during an additive manufacturing process.

[0125] Computing device **12** may receive data indicative of one or more parameters of melt pool **32** from melt pool monitor **15** (**606**). For example, melt pool monitor may be a camera which captures imaging data (e.g., visual and/or thermal imaging data) of build surface **28** and transmit the captured data to computing device **12**. Computing device **12** may determine a current position of melt pool **32** based on the received data (**608**). The determined position of melt pool **32** may be relative to deposition head **54** or absolutely within system **10**.

[0126] Computing device **12** may determine a desired size of melt pool **32** based on the determined position of melt pool **32** (**610**). In some examples, determining a desired size of melt pool **32** may include storing, by computing device **12**, a model of the desired final additively-manufactured component **22**. The model of desired final additively-manufactured component **22** may include desired melt pool sizes at each of a plurality of cells within a matrix. Computing device **12** may compare the determined position of melt pool **32** to the model of the desired final additively-manufactured component to the model of the desired final additively-manufactured component **22**. Computing device **12** may determine the desired size of melt pool **32** based at least partially on the location of the melt pool within the model of the desired final additively-manufactured component **22**.

[0127] In some examples, the desired size of melt pool **32** may be determined based on a sensed edge of component **22**. With reference to FIGS. **9A** and **9B**, in some examples computing device **12** may receive from melt pool monitor **15** data indicative of the geometry of build surface **428**. Computing device may determine at least one boundary of build surface **428**, which in the illustrated example of FIGS. **9A** and **9B** is edge **472** based on the received data, and computing device **12** may determine the desired size of melt pool **32** based on the determined boundary.

[0128] In some examples, computing device **12** may determine the desired size of melt pool **32** based at least partially on topological data of build surface **28** captured by topology sensor **48**. With concurrent reference to FIGS. **10A-10C**, computing device **12** may receive from topology sensor **48**, data indicative of the topology of build surface **528** in the area of the melt pool. Computing device **12** may identify deviation **580** in build surface **528**, and determine the desired size of melt pool **532B** based on the identified deviation.

[0129] Computing device **12** may control energy delivery device **16** based on the determined desired size of melt pool **32** (**612**). Computing device **12** may determine whether size of melt pool **32** meets a threshold for matching the desired size of melt pool **32**. Responsive to determining that the size of melt pool **32** does not meet a threshold for matching the desired melt pool size, computing device **12** may adjust the size of melt pool **32** to meet the threshold for matching the desired size of melt pool **32** by modifying energy delivery device **16**. In some examples, energy delivery device **16** is a laser, and modifying energy delivery device **16** may include modifying a focus of the laser or modifying a working distance **WD** of the laser. In some examples, computing device **12** may maintain a power density of energy delivery device **16** by modifying a power of the laser proportionally to the adjustment of the size of melt pool **32**.

[0130] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware, or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic

circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0131] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various techniques described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware, firmware, or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware, firmware, or software components, or integrated within common or separate hardware, firmware, or software components.

[0132] The techniques described in this disclosure may also be embodied or encoded in an article of manufacture including a computer-readable storage medium encoded with instructions. Instructions embedded or encoded in an article of manufacture including a computer-readable storage medium encoded, may cause one or more programmable processors, or other processors, to implement one or more of the techniques described herein, such as when instructions included or encoded in the computer-readable storage medium are executed by the one or more processors. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a compact disc ROM (CD-ROM), a floppy disk, a cassette, magnetic media, optical media, or other computer readable media. In some examples, an article of manufacture may include one or more computer-readable storage media.

[0133] In some examples, a computer-readable storage medium may include a non-transitory medium. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).

[0134] Various examples have been described. These and other examples are within the scope of the following examples and claims.

[0135] Example 1: An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of an additively-manufactured component being manufactured to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a melt pool monitor configured to observe the melt pool; and a computing device configured to: receive, from the melt pool monitor, data indicative of one or more parameters of the melt pool; determine, based on the received data, a current position of the melt pool; determine, based on the current position of the melt pool, a desired size of the melt pool; and control, based on the desired size of the melt pool, the energy delivery device to form the melt pool of the desired size in the build surface of the component.

[0136] Example 2: The additive manufacturing system of example 1, wherein, to determine the desired size of the melt pool, the computing device is configured to: store a model of a fully formed version of the additively-manufactured component, wherein the model of the fully formed additively-manufactured component includes desired melt pool sizes at each of a plurality of cells within a matrix, compare the position of the melt pool to the model of the desired final additively-manufactured component, determine the position of melt pool within the model of the desired final additively-manufactured component, and determine the desired size of the melt pool based at least partially on the position of the melt pool within the model of the desired final additively-manufactured component.

[0137] Example 3: The additive manufacturing system of any of examples 1 and 2, wherein, to

determine the desired size of the melt pool, the computing device is configured to: receive, from the melt pool monitor, data indicative of a geometry of the build surface, determine at least one boundary of the build surface based on the received data from the melt pool monitor, and determine the desired size of the melt pool based at least partially on the determined at least one boundary of the build surface.

[0138] Example 4: The additive manufacturing system of any of examples 1 through 3, wherein, to determine the desired size of the melt pool, the computing device is configured to: receive, from a topology sensor, data indicative of the topology of the build surface in area vicinity of melt pool.

[0139] Example 5: The additive manufacturing system of example 4, wherein the topology sensor comprises a laser profilometer.

[0140] Example 6: The additive manufacturing system of any of examples 1 through 5, wherein the energy delivery device is a laser configured to deliver energy as an energy beam, wherein the energy beam is configured to impinge upon the build surface to form the melt pool in the build surface, and wherein a size of the melt pool is defined by the greatest dimension of the melt pool in a plane parallel to the build surface.

[0141] Example 7: The additive manufacturing system of any of examples 1 through 6, wherein, to control the energy delivery device to form the melt pool of the desired size in the build surface of the component, the computing device is configured to: determine, based on the received data, a current size of the melt pool; determine whether the current size of the melt pool meets a threshold for matching the desired size of the melt pool; and responsive to determining that the current size of the melt pool does not meet the threshold for matching the desired size of the melt pool, modifying the energy delivery device to form the melt pool with the desired size.

[0142] Example 8: The additive manufacturing system of example 7, wherein the energy delivery device is a laser, and to modify the energy delivery device, the computing device is configured to modify a focus of the laser or modify a working distance of the laser.

[0143] Example 9: The additive manufacturing system of example 8, wherein the computing device is configured to maintain, while modifying the energy delivery device, a power density of the energy delivery device by modifying a power of the laser proportionally to adjustment of the size of the melt pool.

[0144] Example 10: The additive manufacturing system of any of examples 1 through 9, wherein: the powder delivery device and the energy delivery device are parts of a deposition head, and the energy delivery device and the powder delivery device are independently controllable.

[0145] Example 11: The additive manufacturing system of example 10, wherein: the powder delivery device comprises one or more delivery nozzles from which the powder stream is directed, and to control the powder delivery device, the computing device is configured to control a position of the of the one or more delivery nozzles.

[0146] Example 12: The additive manufacturing system of example 11, wherein: individual portions of the powder stream are directed toward the melt pool from a plurality of delivery nozzles, the individual portions of the powder stream are configured to converge at a convergence point, and the computing device is configured to modify the convergence point by adjusting the powder delivery device in response to a change in the size of the melt pool.

[0147] Example 13: The additive manufacturing system of any of examples 1 through 12, further comprising the additively-manufactured component, wherein the additively manufactured component is a gas turbine engine component.

[0148] Example 14: A method includes delivering, via an energy delivery device, energy to a build surface of an additively-manufactured component being manufactured to form a melt pool in the build surface of the component; delivering, via a powder delivery device, a powder stream to the melt pool to add material to the component; receiving, from a melt pool monitor, data indicative of one or more parameters of the melt pool; determining, via a computing device, based on the received data, a current position of the melt pool; determining, via the computing device, based on

the current position of the melt pool, a desired size of the melt pool; and controlling, via the computing device, based on the desired size of the melt pool, the energy delivery device to form the melt pool of the desired size in the build surface of the component.

[0149] Example 15: The method of example 14, further includes storing, by the computing device, a model of a fully formed version of the additively-manufactured component, wherein the model of a fully formed version of the additively-manufactured component includes desired melt pool sizes at each of a plurality of cells within a matrix; comparing, by the computing device, the position of the melt pool to the model of the desired final additively-manufactured component; determining, by the computing device, the position of melt pool within the model of the desired final additively-manufactured component; and determine the desired size of the melt pool based at least partially on the position of the melt pool within the model of the desired final additively-manufactured component.

[0150] Example 16: The method of any of examples 14 and 15, further includes receiving, from the melt pool monitor, data indicative of a geometry of the build surface, determining, by the computing device, at least one boundary of the build surface based on the received data from the melt pool monitor, and determining, by the computing device, the desired size of the melt pool based at least partially on the determined at least one boundary of the build surface.

[0151] Example 17: The method of any of examples 14 through 16, further includes receiving, from a topology sensor, data indicative of the topology of the build surface in an area of melt pool.

[0152] Example 18: The method of any of examples 14 through 17, further includes determining, by the computing device, whether the size of the melt pool meets a threshold for matching the desired size of the melt pool, and responsive to determining that the size of the melt pool does not meet a threshold for matching the desired melt pool size, adjusting, by the computing device, the size of the melt pool to meet the threshold for matching the desired size of the melt pool by modifying the energy delivery device.

[0153] Example 19: The method of example 18, wherein the energy delivery device is a laser, and wherein modifying the energy delivery device comprises modifying a focus of the laser or modifying a working distance of the laser.

[0154] Example 20: The method of example 19, further comprising maintaining a power density of the energy delivery device by modifying a power of the laser proportionally to the adjustment of the size of the melt pool.

Claims

1. An additive manufacturing system comprising: an energy delivery device configured to deliver energy to a build surface of an additively-manufactured component being manufactured to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a melt pool monitor configured to observe the melt pool; and a computing device configured to: receive, from the melt pool monitor, data indicative of one or more parameters of the melt pool; determine, based on the received data, a current position of the melt pool; determine, based on the current position of the melt pool, a desired size of the melt pool; and control, based on the desired size of the melt pool, the energy delivery device to form the melt pool of the desired size in the build surface of the component.

2. The additive manufacturing system of claim 1, wherein, to determine the desired size of the melt pool, the computing device is configured to: store a model of a fully formed version of the additively-manufactured component, wherein the model of the fully formed additively-manufactured component includes desired melt pool sizes at each of a plurality of cells within a matrix, compare the position of the melt pool to the model of the desired final additively-manufactured component, determine the position of melt pool within the model of the desired final additively-manufactured component, and determine the desired size of the melt pool based at least

partially on the position of the melt pool within the model of the desired final additively-manufactured component.

3. The additive manufacturing system of claim 1, wherein, to determine the desired size of the melt pool, the computing device is configured to: receive, from the melt pool monitor, data indicative of a geometry of the build surface, determine at least one boundary of the build surface based on the received data from the melt pool monitor, and determine the desired size of the melt pool based at least partially on the determined at least one boundary of the build surface.

4. The additive manufacturing system of claim 1, wherein, to determine the desired size of the melt pool, the computing device is configured to: receive, from a topology sensor, data indicative of the topology of the build surface in area vicinity of melt pool; identify, based on the received data from the topology sensor, a deviation in the build surface, and determine, based on the identified deviation in the build surface, the desired size of the melt pool.

5. The additive manufacturing system of claim 4, wherein the topology sensor comprises a laser profilometer.

6. The additive manufacturing system of claim 1, wherein the energy delivery device is a laser configured to deliver energy as an energy beam, wherein the energy beam is configured to impinge upon the build surface to form the melt pool in the build surface, and wherein a size of the melt pool is defined by the greatest dimension of the melt pool in a plane parallel to the build surface.

7. The additive manufacturing system of claim 1, wherein, to control the energy delivery device to form the melt pool of the desired size in the build surface of the component, the computing device is configured to: determine, based on the received data, a current size of the melt pool; determine whether the current size of the melt pool meets a threshold for matching the desired size of the melt pool; and responsive to determining that the current size of the melt pool does not meet the threshold for matching the desired size of the melt pool, modifying the energy delivery device to form the melt pool with the desired size.

8. The additive manufacturing system of claim 7, wherein the energy delivery device is a laser, and to modify the energy delivery device, the computing device is configured to modify a focus of the laser or modify a working distance of the laser.

9. The additive manufacturing system of claim 8, wherein the computing device is configured to maintain, while modifying the energy delivery device, a power density of the energy delivery device by modifying a power of the laser proportionally to adjustment of the size of the melt pool.

10. The additive manufacturing system of claim 1, wherein: the powder delivery device and the energy delivery device are parts of a deposition head, and the energy delivery device and the powder delivery device are independently controllable.

11. The additive manufacturing system of claim 10, wherein: the powder delivery device comprises one or more delivery nozzles from which the powder stream is directed, and to control the powder delivery device, the computing device is configured to control a position of the of the one or more delivery nozzles.

12. The additive manufacturing system of claim 11, wherein: individual portions of the powder stream are directed toward the melt pool from a plurality of delivery nozzles, the individual portions of the powder stream are configured to converge at a convergence point, and the computing device is configured to modify the convergence point by adjusting the powder delivery device in response to a change in the size of the melt pool.

13. The additive manufacturing system of claim 1, further comprising the additively-manufactured component, wherein the additively manufactured component is a gas turbine engine component.

14. A method comprising: delivering, via an energy delivery device, energy to a build surface of an additively-manufactured component being manufactured to form a melt pool in the build surface of the component; delivering, via a powder delivery device, a powder stream to the melt pool to add material to the component; receiving, from a melt pool monitor, data indicative of one or more parameters of the melt pool; determining, via a computing device, based on the received data, a

current position of the melt pool; determining, via the computing device, based on the current position of the melt pool, a desired size of the melt pool; and controlling, via the computing device, based on the desired size of the melt pool, the energy delivery device to form the melt pool of the desired size in the build surface of the component.

15. The method of claim 14, further comprising: storing, by the computing device, a model of a fully formed version of the additively-manufactured component, wherein the model of a fully formed version of the additively-manufactured component includes desired melt pool sizes at each of a plurality of cells within a matrix; comparing, by the computing device, the position of the melt pool to the model of the desired final additively-manufactured component; determining, by the computing device, the position of melt pool within the model of the desired final additively-manufactured component; and determine the desired size of the melt pool based at least partially on the position of the melt pool within the model of the desired final additively-manufactured component.

16. The method of claim 14, further comprising: receiving, from the melt pool monitor, data indicative of a geometry of the build surface, determining, by the computing device, at least one boundary of the build surface based on the received data from the melt pool monitor, and determining, by the computing device, the desired size of the melt pool based at least partially on the determined at least one boundary of the build surface.

17. The method of claim 14, further comprising: receiving, from a topology sensor, data indicative of the topology of the build surface in an area of melt pool, identifying, by the computing device, based on the received data from the topology sensor, a deviation in the build surface, and determining, based on the identified deviation in the build surface, the desired size of the melt pool.

18. The method of claim 14, further comprising: determining, by the computing device, whether the size of the melt pool meets a threshold for matching the desired size of the melt pool, and responsive to determining that the size of the melt pool does not meet a threshold for matching the desired melt pool size, adjusting, by the computing device, the size of the melt pool to meet the threshold for matching the desired size of the melt pool by modifying the energy delivery device.

19. The method of claim 18, wherein the energy delivery device is a laser, and wherein modifying the energy delivery device comprises modifying a focus of the laser or modifying a working distance of the laser.

20. The method of claim 19, further comprising maintaining a power density of the energy delivery device by modifying a power of the laser proportionally to the adjustment of the size of the melt pool.
